



Introducing

Geriatrics



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LECTURE OBJECTIVES

- Understand Current Concepts in Geriatrics
- Know Key Factors that may result in Altered Presentation in Geriatric Patients
- Define FRAILITY and SARCOPENIA
- Be aware of the Geriatric Syndromes
- Understand the Pharmacokinetic and Pharmacodynamic changes associated with aging
- Define a “Prescribing Cascade”



LEARNING OBJECTIVES, continued

- List 4 types of Drug Interactions
- Identify strategies for an effective approach to elder visits
- Be able to utilize the tools necessary to assess the cognitive, mental health, physical limitations and fall risks in the evaluation and treatment of the geriatric patient
- Be able to list aspects of aging related to appropriate OMM
- Define Immunosenesence
- Discuss the importance of and rationale for vaccines in the elderly

THE AGING WORLD

- Not only has world population increased, but the percentage of older adults will continue to increase to historic levels
- Average life expectancy in US increased from 47 years in 1900 to nearly 76.1 years in 2021
- This shift in population age has implications for current and upcoming generations of physicians



WHAT IS THE RELEVANCE OF GERIATRICS?

- The aging population is relevant to most specialties, especially primary care
- Familiarity/expertise with geriatric issues will be essential, likely professionally and/or personally
- Developing skills with geriatrics has good crossover relevance to other areas within healthcare



What is the relevance of Geriatrics?

- Heart disease, cancer, and stroke have become the leading causes of death among older adults
- Adults surviving into late life suffer from high rates of chronic illness; 80% have at least one and 50% have at least 2 chronic conditions
- Strong association between the presence of geriatric syndromes and dependency in the activities of daily living (ADL's).



What is the relevance of Geriatrics?

- Decline in function and loss of independence are NOT an inevitable consequence of aging
- Given the high prevalence and impact of chronic health problems among older patients, evidence-based interventions to address these problems becomes increasingly important to maximize both the quantity and quality of life for older adults.



SKILLS USEFUL IN GERIATRICS ARE ALSO HELPFUL FOR

- Assessment related to disabilities from injuries or disease
- Wounded Warriors
- Neurological Conditions
- Rehabilitation Settings



1
Diseases and
Aging Occur
on a
Continuum

2
Disease is
Multifactorial

3
Atypical
Presentations

50% of all Americans 65+ have some disability

63% of 85+ can't manage ADLs

47% 85 + have dementia

33% 60 + affected by falls

Hospital admits due to adverse drug
reaction 6X greater in elderly than
general Population



YOU DON'T WAKE UP OLD ONE DAY

Aging is a metabolic functional and *normal* process

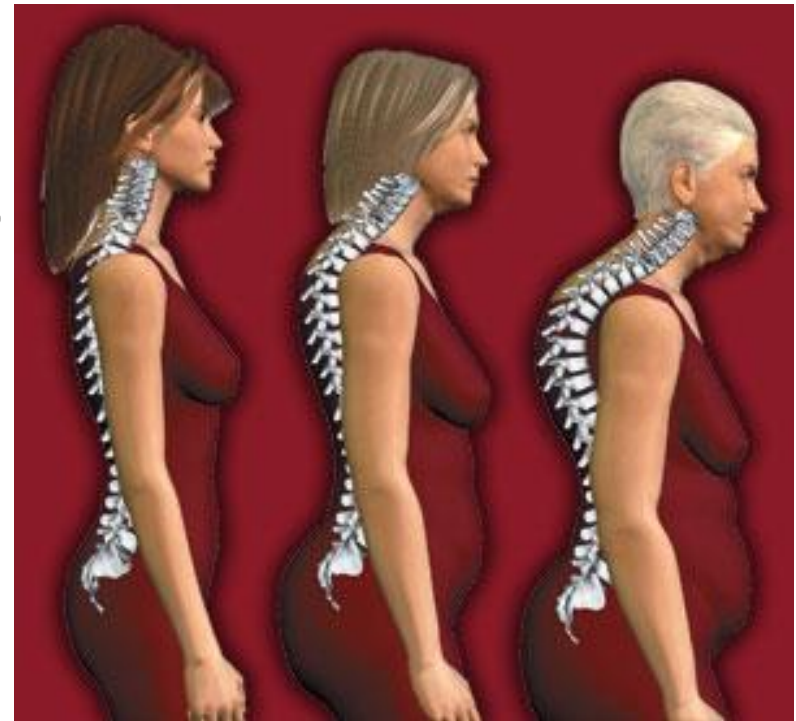


Why Do We Age?

Errors occur

Through wear and tear on
cells

Through DNA protein cross linking



Programmed Senescence Theory

Sequential switching on and off of genes is in DNA

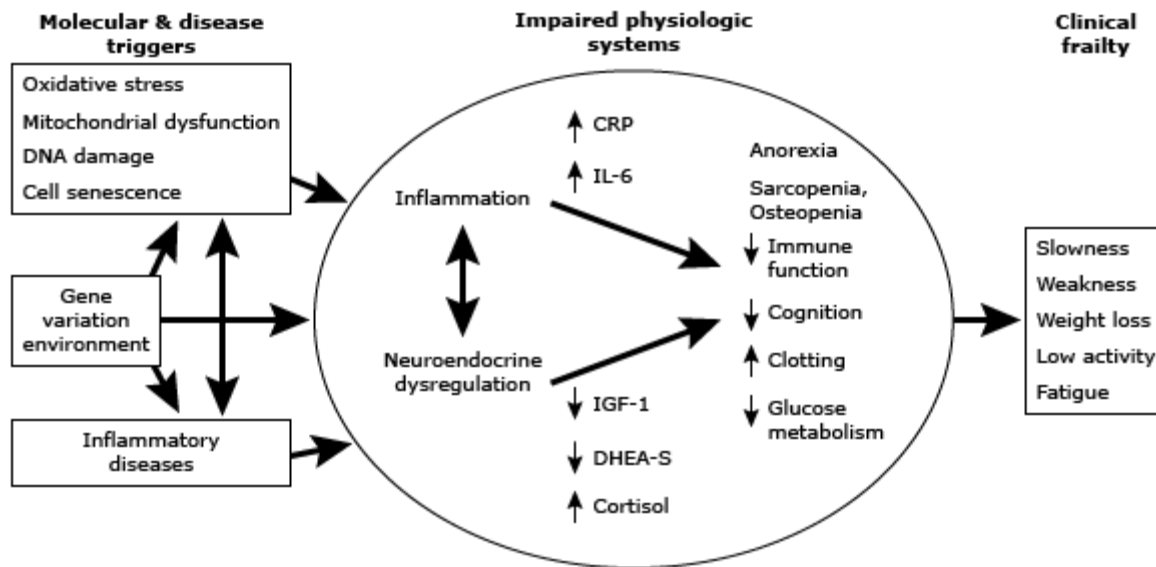
Free Radicals Damage Protein and DNA

Normal byproducts of cellular activity

Chronic Pro-inflammatory State

Leads to decreased physiologic reserve





Decreased Physiologic Reserve

- » Examples

- » Decreased muscle mass
- » Decreased respiratory function
- » **Decreased immune response**

- » Result

- » Increased vulnerability to decreased mobility
- » Increased vulnerability to temperature changes
- » Increased vulnerability to infection
- » Increased likelihood of developing complex geriatric syndromes

Age Bands

Frisky

65-74 years

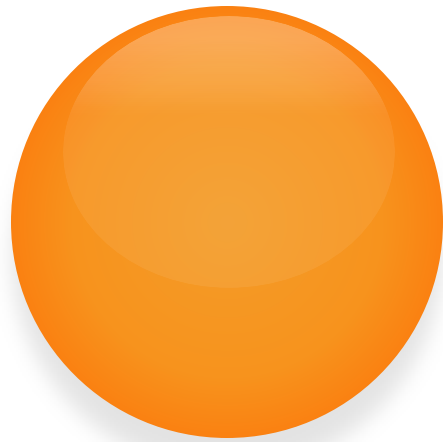
Frail

75 - 84 years

Fragile

85+



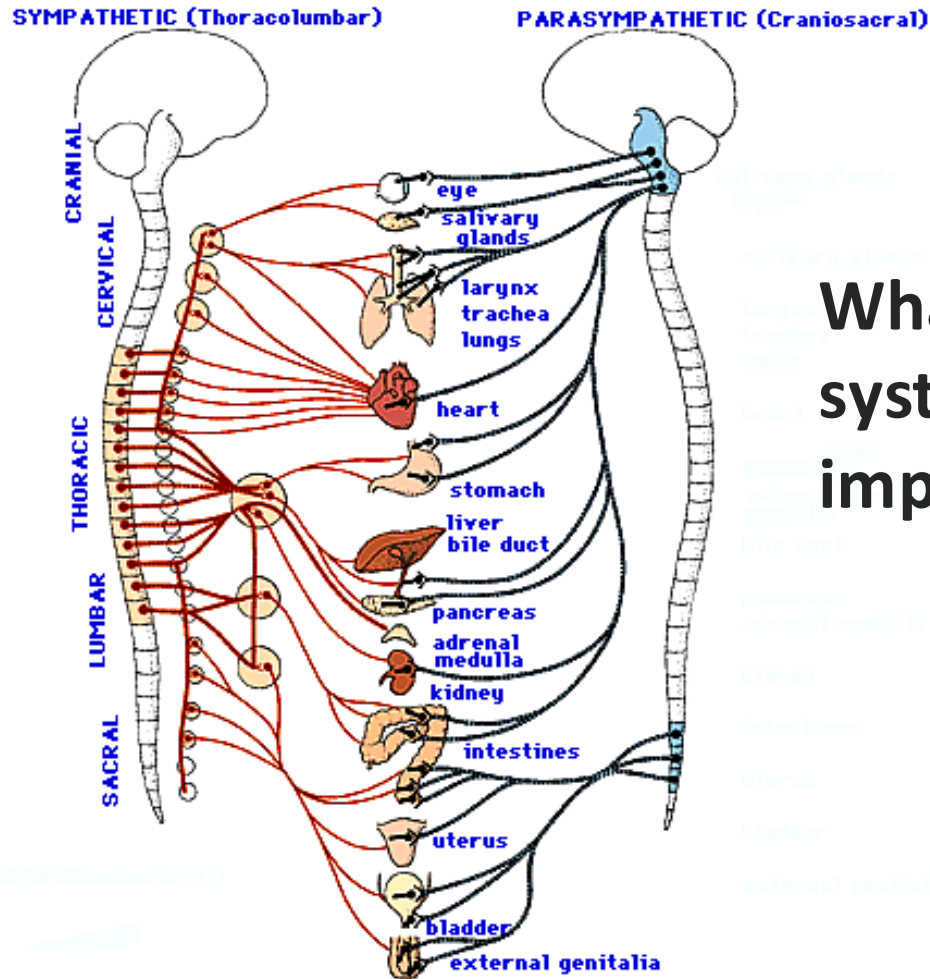


DISEASE IS MULTIFACTORIAL

THE SECOND CONCEPT



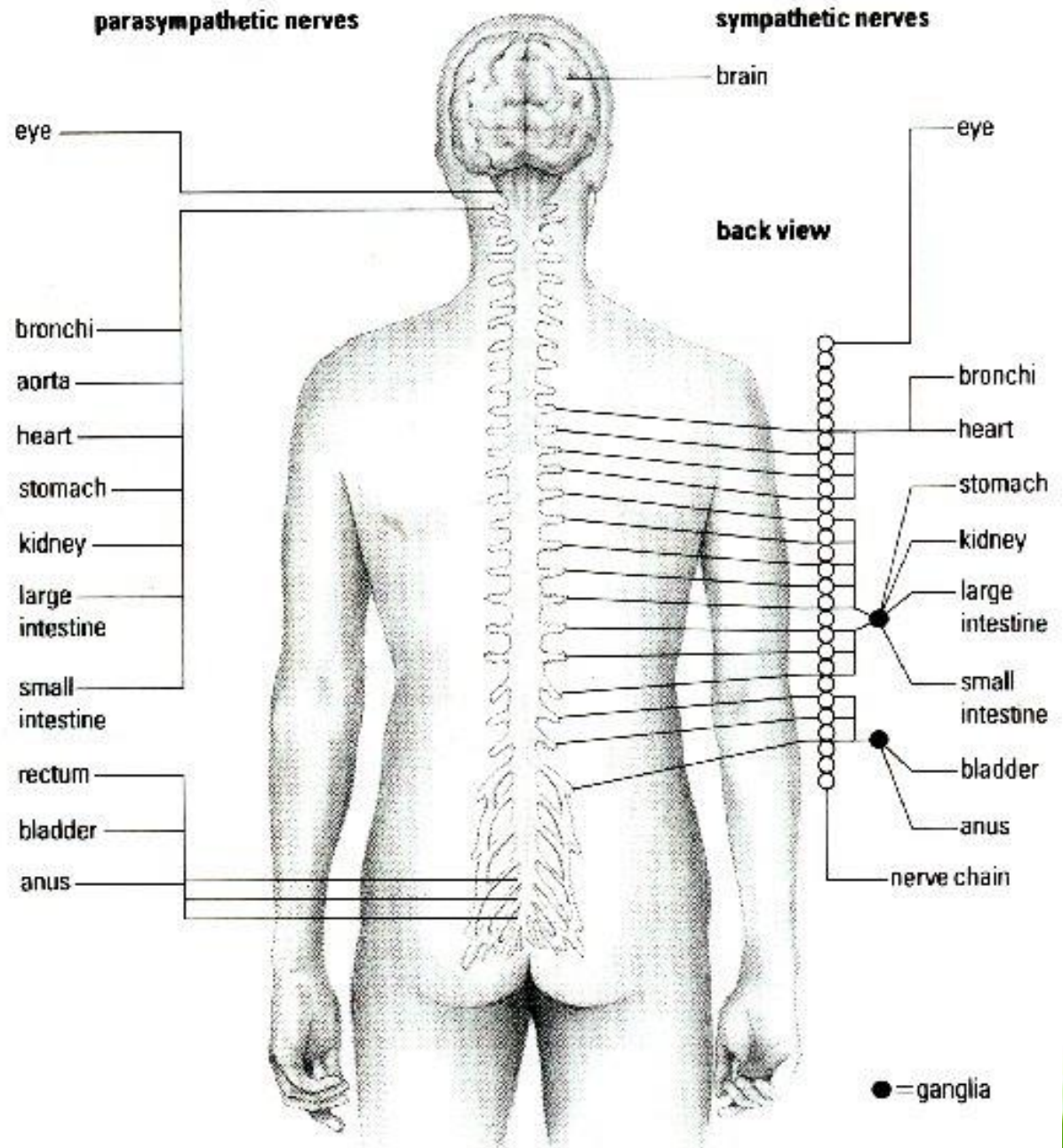
THE AUTONOMIC NERVOUS SYSTEM



What body part, organ system or disease isn't impacted by the ANS?

THE AUTONOMIC NERVOUS SYSTEM

Anxiety
Arrhythmia
Depression
Asthma
HTN
CAD
Orthostatic HTN
Ataxia
Urinary Tract Infection
Palpitations



Biomechanical Explanation

Viscero-somatic Reflexes

- Visceral dysfunction can cause reflex sympathetic activity in the paravertebral musculature of the associated spinal level

Somato-visceral reflexes

- somatic dysfunction can cause reflex sympathetic activity in the viscera



STRUCTURE GOVERNS FUNCTION

- Physical Impairment causes difficulty in function which leads to reduced participation in self care and society.





Atypical Presentations

It doesn't necessarily walk like a duck or quack like one.....

Key factors in Atypical Presentation

1. Under reporting
2. Aging *and* Disease affect Signs and Symptoms
3. Geriatric syndromes

Under Reporting

- Patient, family and doctor dismiss treatable symptoms and impairments as age-related changes
- Denial, from fear of economic, social, or functional consequences
- Cognitive impairment and depressive symptoms limit the ability or desire



Aging *and* disease affect signs and symptoms

- Atypical Presentation
 - ABSENCE OF FEVER
 - PAIN
- Age related changes & Co-existing conditions alter presentation
 - One condition may mask another
 - Decreased perception and sensory function may alter presentation (due to aging or disease)
 - Most commonly happens in fragile and frail



ATYPICAL PRESENTATIONS

- A patient may present with ACS and minimal pain due to decreased sensory function
- A patient with dementia who lacks the capacity to communicate symptoms



FEVER

- **ABSENT IN 30-50% OF FRAIL, ELDERLY ADULTS EVEN IN THE SETTING OF SERIOUS INFECTION**
SUCH AS PNEUMONIA OR ENDOCARDITIS
- **THE BLUNTED FEBRILE RESPONSE IS DUE TO CHANGES IN MULTIPLE SYSTEMS RESPONSIBLE FOR THERMOREGULATION:** Shivering, vasoconstriction, hypothalamic regulation and thermogenesis by brown adipose tissue, are all impaired with advanced age

Examples of some diseases that may present **differently** in the elderly

- **PNEUMONIA**- Patient may be afebrile, delirious or AMS, and dehydrated
- **ACUTE CORONARY SYNDROME (ACS)**- patient may have altered pain sensation, syncope or fall as presenting C/O
- **Acute UTI** patient may present with confusion, delirium, agitation and dehydration



One *Common*, ATYPICAL CASE

- 88 y/o F with dementia, chronic Afib; BIB Paramedics to ED from SNF; RN reported increasing agitation, anorexia over previous 3 days. Sent to ED today due to HR 140 AND AMS
- PERTINENT + FINDINGS:
- VS 97.2 -144-18-100/70 O2sat 95%
- Gen appearance: anxious, agitated, oriented to name only



Case continued

- PE- oral- dry mucous membranes
- CV tachy, irregularly irregular rate
- Skin tenting present (sign of dehydration)
- STUDIES EKG AFIB WITH RVR 144
- CBC WBC 4.0 WITH LEFT SHIFT (INCREASED BANDS)
- UA FULL FIELD WBC'S

Geriatric Syndromes



- Frailty- the hallmark geriatric syndrome
- Sarcopenia
- Incontinence
- Dementia and Depression
- Falls
- Dizziness
- Syncope
- Pressure Ulcers
- Elder Mistreatment or Abuse



FRAILTY

- A FORERUNNER TO OTHER GERIATRIC SYNDROMES
- DEFINITION-A syndrome of physiological decline in late life, characterized by marked vulnerability to adverse health outcomes, including procedural complications, institutionalization, disability, and death.



Epidemiology studies of frailty

- Factors associated with an increased prevalence of frailty:
- Older age Lower educational level
- Current smoker Not married
- Current use of postmenopausal HRT (hormone replacement therapy)
- In US sample, AA or Hispanic ethnicity
- Depression



THE FRAIL SCALE

- *Fatigue (Are you fatigued?)*
- *Resistance (Can you climb one flight of stairs?)*
- *Ambulation (Can you walk one block?)*
- *Illnesses (greater than five)*
- *Loss of weight (greater than 5%) past 3 months*
- *Positive response to 3 or more indicates frailty*
- *Positive response to one or two indicates pre-frailty*

The Study of Osteoporotic Fractures

- The SOF frailty tool defines frailty as the presence of at least two of three components:
- **Weight loss of 5 % in the last year**
- **Inability to rise from a chair five times without use of arms or**
- **A “no” response to the question “Do you feel full of energy?”**



FRAILITY AND COGNITIVE IMPAIRMENT

- Frailty is associated with an **increased risk of** cognitive impairment and an **increased rate** of cognitive decline with aging
- **Conversely**, the **presence of cognitive impairment** **increases** the likelihood of adverse health outcomes in geriatric patients who meet criteria for physical frailty, and **hence may be considered an additive factor** to frailty in those older adults with both conditions



Sarcopenia



a syndrome characterized by the loss of muscle mass, strength, and performance

Sarcopenia is associated with increased rates of functional impairment, disability, falls and mortality

Causes of **Sarcopenia** can include endocrine changes:

1. insulin resistance, which increases with age
- 2 Reductions in testosterone and estrogen that accompany aging appear to accelerate the development of **sarcopenia**.

Other changes include:

3. Activation of proinflammatory catabolic cytokines such as IL-1 and IL-6
4. Decreased physical activity
5. Suboptimal protein intake



MUAC

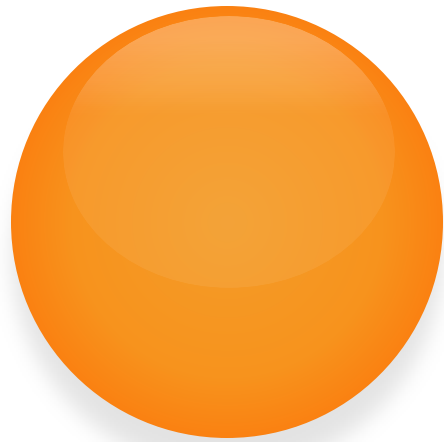
- Body fat and lean muscle mass may be measured using anthropometric measurements such as **mean upper arm circumference (MUAC)**
- MUAC measures the circumference of the left upper arm at the mid-point between the tip of the shoulder and the tip of the elbow (olecranon to acromium)
- MUAC less than 22cm in women and 23 cm in men suggests chronic energy deficiency



FRAILITY: DIFFERENTIAL DIAGNOSES TO CONSIDER

- *DEPRESSION*
- *MALIGNANCY*-Lymphoma, multiple myeloma, occult solid tumors
- *RHEUMATOLOGIC DISEASE*- Polymyalgia rheumatica, vasculitis
- *ENDOCRINOLOGIC DISEASE*- Hyper-or hypothyroidism, DM
- *CARDIOVASCULAR DISEASE*-HTN, HF, CAD, and PVD
- *RENAL DISEASE*-Renal insufficiency
- *HEMATOLOGIC DISEASE*-Myelodysplasia, Iron deficiency, and Pernicious anemia
- *NUTRITIONAL DEFICITS*-Vitamin deficiencies
- *NEUROLOGIC DISEASE*-Parkinson's, vascular dementia, and serial lacunar infarcts





URINARY INCONTINENCE

A COMMON GERIATRIC SYNDROME



URGE INCONTINENCE

- **URGE INCONTINENCE**-DAYTIME FREQUENCY, NOCTURIA, BOTHERSOME URGENCY
- FIRST-LINE **THERAPY IS BLADDER TRAINING**
- SECOND-LINE TX IS ANTI-CHOLINERGIC DRUGS
- OXYBUTININ, TOLTERODINE
- MUST CONSIDER THE side effects OF THESE DRUGS FIRST!!



STRESS INCONTINENCE

- INVOLUNTARY RELEASE OF URINE SECONDARY TO EFFORT OR EXERTION
- EX. SNEEZING, COUGHING
- FIRST LINE-THERAPY IS PELVIC FLOOR MUSCLE TRAINING FOR WOMEN

FUNCTIONAL INCONTINENCE

- UNABLE TO GET TO THE BATHROOM IN TIME BECAUSE OF MENTAL OR PHYSICAL LIMITATIONS
- REGULAR PROMPTED VOIDING WITH ASSISTANCE TO COMMUNE OR PORTABLE COMMUNE AND TX UNDERLYING DISORDERS

OSTEOPATHIC CONSIDERATIONS IN INCONTINENCE



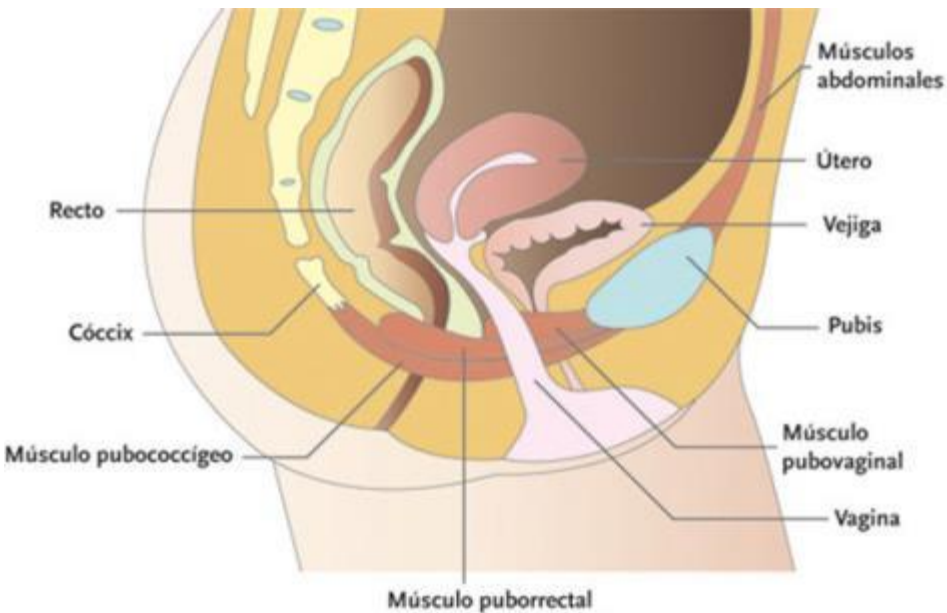
- Autonomic
 - SNS from T10-T12, hypogastric nerve, pelvic plexus
 - PSNS from S2-S4, pelvic nerve, pelvic plexus
 - SNS inhibition of detrusor contraction and PSNS activation allow for storage
 - PSNS stimulation relaxes external urethral sphincters and decreases pelvic floor muscle contraction

OSTEOPATHIC CONSIDERATIONS IN INCONTINENCE



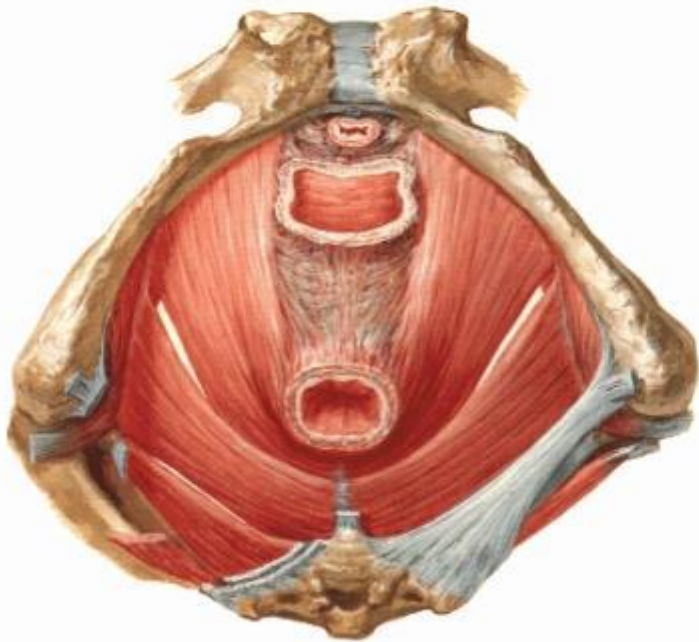
- Biomechanical
 - Somatic dysfunction of the pelvis can contribute to urinary incontinence
 - Treat with Mitchell model and or MFR/BLT
 - Muscle imbalance in the pelvic floor can affect bladder and urethral orientation
 - Treat with MFR, muscle inhibition, respiratory assist, visceral manipulation
 - If teaching Keigel exercises, do not have patients actually “hold their urine”

OSTEOPATHIC CONSIDERATIONS IN INCONTINENCE

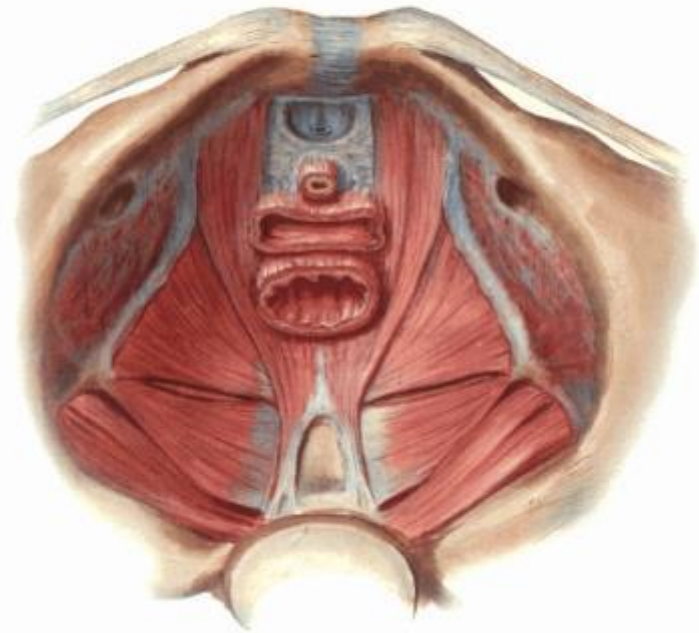


OSTEOPATHIC CONSIDERATIONS IN INCONTINENCE

Pelvic Diaphragm of Female
Inferior View



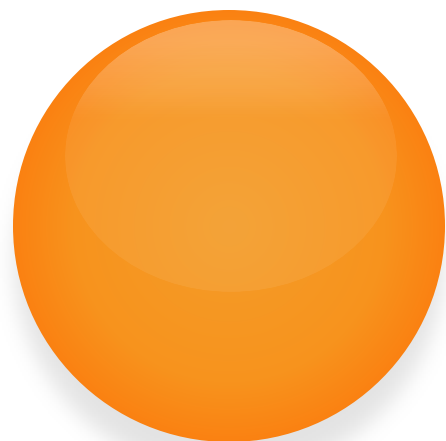
Pelvic Diaphragm of Female
Superior View



OSTEOPATHIC CONSIDERATIONS IN INCONTINENCE



- Circulation
 - Internal iliac vessels
 - Common iliac lymphatics
- Screening
 - Zink, fascial/fluid wave, leg tug
 - Cranial strain pattern(s)



PRESSURE ULCERS

GERIATRIC SYNDROME

PRESSURE ULCERS

- ISCHEMIC SOFT TISSUE INJURIES RESULTING FROM PRESSURE, USUALLY OVER BONY PROMINENCES
- THE EXTERNAL APPEARANCE OF A PRESSURE ULCER MAY UNDERESTIMATE THE EXTENT OF INJURY
- WOUND CARE REFERRAL NEEDED

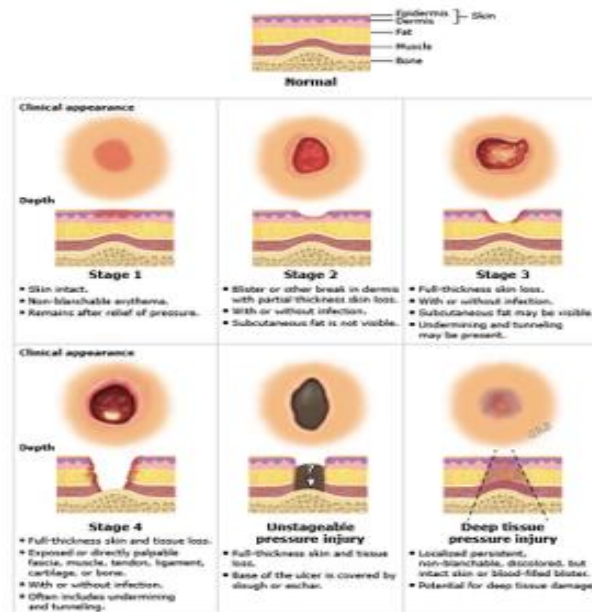
ULCER STAGE

- **STAGE 1:** THE SKIN IS INTACT WITH NON-BLANCHABLE REDNESS
- **STAGE 2:** SHALLOW ULCER WITH A RED-PINK WOUND BED OR SERUM-FILLED BLISTER
- **STAGE 3:** SUBCUTANEOUS FAT MAY BE VISIBLE
- **STAGE 4:** EXPOSED BONE, TENDON, OR MUSCLE
- ALWAYS CONSIDER THE POSSIBILITY OF UNDERLYING OSTEOMYELITIS



Characteristics of Pressure-induced Injury by stage

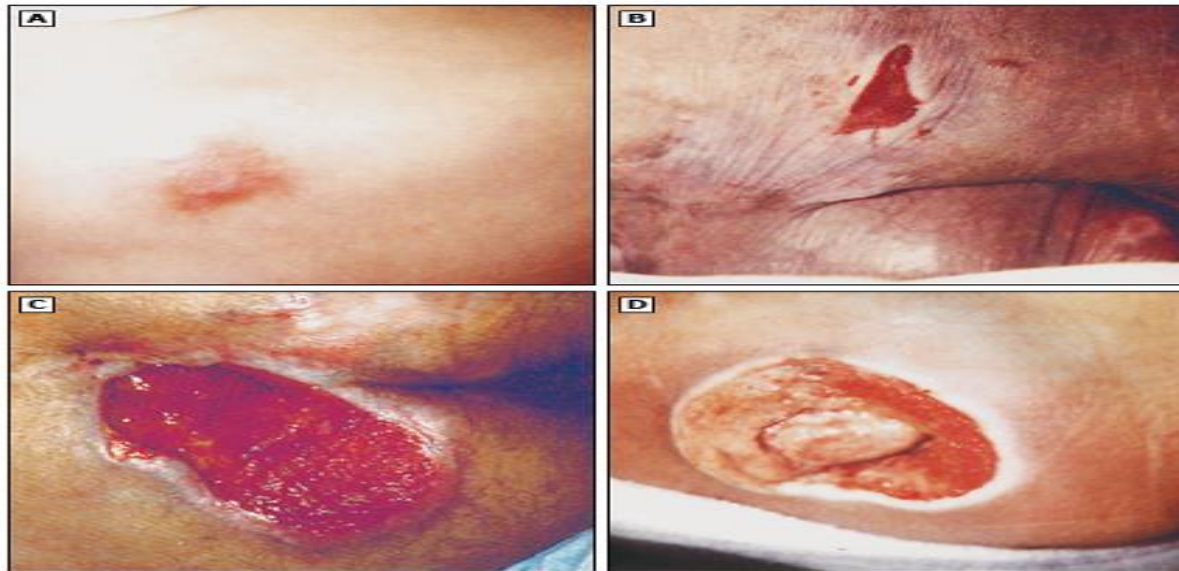
Characteristics of pressure-induced injury by stage



These illustrations highlight the characteristics of the stages of pressure-induced injuries as defined by the National Pressure Ulcer Advisory Panel. Each panel demonstrates the clinical appearance of the involved skin as well as the cross section of the depth of injury. Most pressure-induced injuries are associated with an underlying bony prominence. There is always the potential for deeper tissue damage than appreciated on clinical examination, as illustrated by the panel "Deep tissue pressure injury."

Adapted from definitions by the National Pressure Ulcer Advisory Panel. UpToDate®

Pressure-induced injury by stage



Pressure-induced injury staging^[1].

(A) Stage 1 – Skin intact but with nonblanchable redness.

(B) Stage 2 – Partial-thickness loss of skin with exposed dermis.

(C) Stage 3 – Full-thickness loss of skin, in which adipose (fat) is visible in the ulcer and granulation tissue and epibole (rolled wound edges) are often present.

(D) Stage 4 – Full-thickness skin and tissue loss with exposed or directly palpable fascia, muscle, tendon, ligament, cartilage, or bone in the ulcer.

Not shown:

Unstageable pressure injury – Full-thickness skin and tissue loss in which the extent of tissue damage within the ulcer cannot be confirmed because it is obscured by slough or eschar.

Deep tissue pressure injury – Deep tissue pressure injury should be suspected whenever there is a localized area of skin (intact or non-intact) with persistent nonblanchable deep red, maroon, or purple discoloration or epidermal separation revealing a dark wound bed or blood-filled blister.

Reference:

1. National Pressure Injury Advisory Panel. www.npiap.com.
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TREATMENT OF PRESSURE ULCERS

- AIR-FLUIDIZED BEDS ENHANCE HEALING OF PRESSURE ULCERS COMPARED TO STANDARD HOSPITAL MATTRESSES
- DRESSINGS SHOULD MAINTAIN A MOIST WOUND ENVIRONMENT AND MANAGE EXUDATES
- RESTRICT SYSTEMIC ANTIBX FOR CELLULITIS TX (ERYTHEMA, WARMTH, PAIN)
- DEBRIDE ESCHARS AND NONVIABLE TISSUE



LAB TESTING

- WHY? IN ORDER TO R/O TREATABLE CONDITIONS
- CBC
- Basic Metabolic panel
- LFT's including albumin
- Vitamin B12
- Vitamin D
- TSH



Caring for the elderly



Goals of Care For Older Adults

- Physiologic and functional status vary greatly among older adults
- Treatment decisions, both preventive and therapeutic, should be considered based on individual needs.
- Age alone should not be the sole determinant for many interventions, and all tx should aim to preserve function and maximize qol.



Overview of Prevention for Older Adults

- Healthy lifestyle
- ASA no longer recommended for primary prevention
- Immunizations
- CA screening
- CV screening
- DM screening
- Functional and Psychosocial evaluation
- Osteoporosis
- Vision screening
- Hearing loss
- Nutrition
- Falls and Mobility
- Incontinence

Healthy Lifestyle

- Physical activity - very important to improve strength, flexibility, mobility and fitness
- Develop an individualized “activity plan”
- Tobacco use – important to assess and counsel on how to quit if currently smoking
- Alcohol use – assess to identify patients with alcohol-related problems



Alcohol use in the elderly

- 50% of adults over 65 years consume alcohol
- 14.5% drink more than the recommended weekly allowance (>7 drinks/week) or binge
- Increase risk of falls; may negatively impact function, cognition and general health
- Risk factors for ETOH abuse in elderly: bereavement, depression, anxiety, pain, disability, prior ETOH abuse hx

Aspirin for primary prevention?? Not so fast.....

- Aspirin may decrease the risk for both CV disease and a variety of cancers, **but increases the risk of gastrointestinal bleeding.**
- Individual risks for bleeding, stroke, and coronary events should be reviewed with older patients in the process of shared decision making regarding aspirin use as primary prevention.
- USPSTF recommends against initiating low-dose ASA use for the primary prevention of CVD in adults 60 years or older. Grade D (not recommended)

Cancer Screening

- Clinicians should assess the benefits and risks of screening for older adults on an individual basis, considering the patient's estimated remaining life expectancy, and help patients make decisions based on personal values and preferences.
- Consideration given to colorectal, breast, cervical, and lung cancer screening likely to be most effective in reducing cancer-specific mortality.



Immunosenescence

- Defined as age-related decline in immunity, which significantly contributes to the susceptibility of older adults to serious, vaccine-preventable conditions, including influenza, pneumonia, shingles, and SARS-COV-2.
- Vaccines are an effective way of stimulating and heightening the immune response and boosting waning immunity.



VACCINES IN THE ELDERLY POPULATION

- According to the National Center for Health Statistics, in 2015, among adults 50 years and older, four vaccine-preventable diseases alone-influenza, pneumococcal disease, herpes zoster, and pertussis-cost the USA more than \$26 billion annually.



OLDER ADULT IMMUNIZATIONS

- INFLUENZA- ANNUALLY; High dose available also
- PNEUMOCOCCAL ALL ADULTS => 50 YRS
 - 1. one dose of PCV 20 OR PCV 21 OR
 - 2. one dose of PCV15 followed in 1yr with one dose of PPSV 23
- HERPES ZOSTER ALL ADULTS =>50YRS OF AGE- two-dose recombinant vaccine. 2-6 months after first dose
- 1 dose Tdap, then Td or Tdap booster every 10 years
- Covid-19 Pfizer/Moderna 2 doses initially and Boost with => 2 or more doses of the 2024-2025 vaccine
- RSV-Respiratory Syncytial Virus all adults ages 75 and older and adults ages 60–74 at increased risk of severe RSV.



THE GERIATRIC ASSESSMENT: ANNUALLY AT AGE 75 YEARS

MAJOR COMPONENTS OF THE GERIATRIC ASSESSMENT

- FUNCTIONAL STATUS-ADL/GAIT SPEED
- FALLS/IMBALANCE
- COGNITION
- MOOD DISORDERS
- POLYPHARMACY
- SOCIAL AND FINANCIAL SUPPORT
- GOALS OF CARE
- ADVANCED CARE PREFERENCES

ADVANCED CARE PREFERENCES

- Begin discussions about preferences for specific tx while the patient still has the cognitive capacity to make these decisions
- In-the-moment decision-making, including choosing an appropriate decision-maker to serve as surrogate and
- Think about factors other than patient's stated preferences

DIAGNOSTIC TESTING

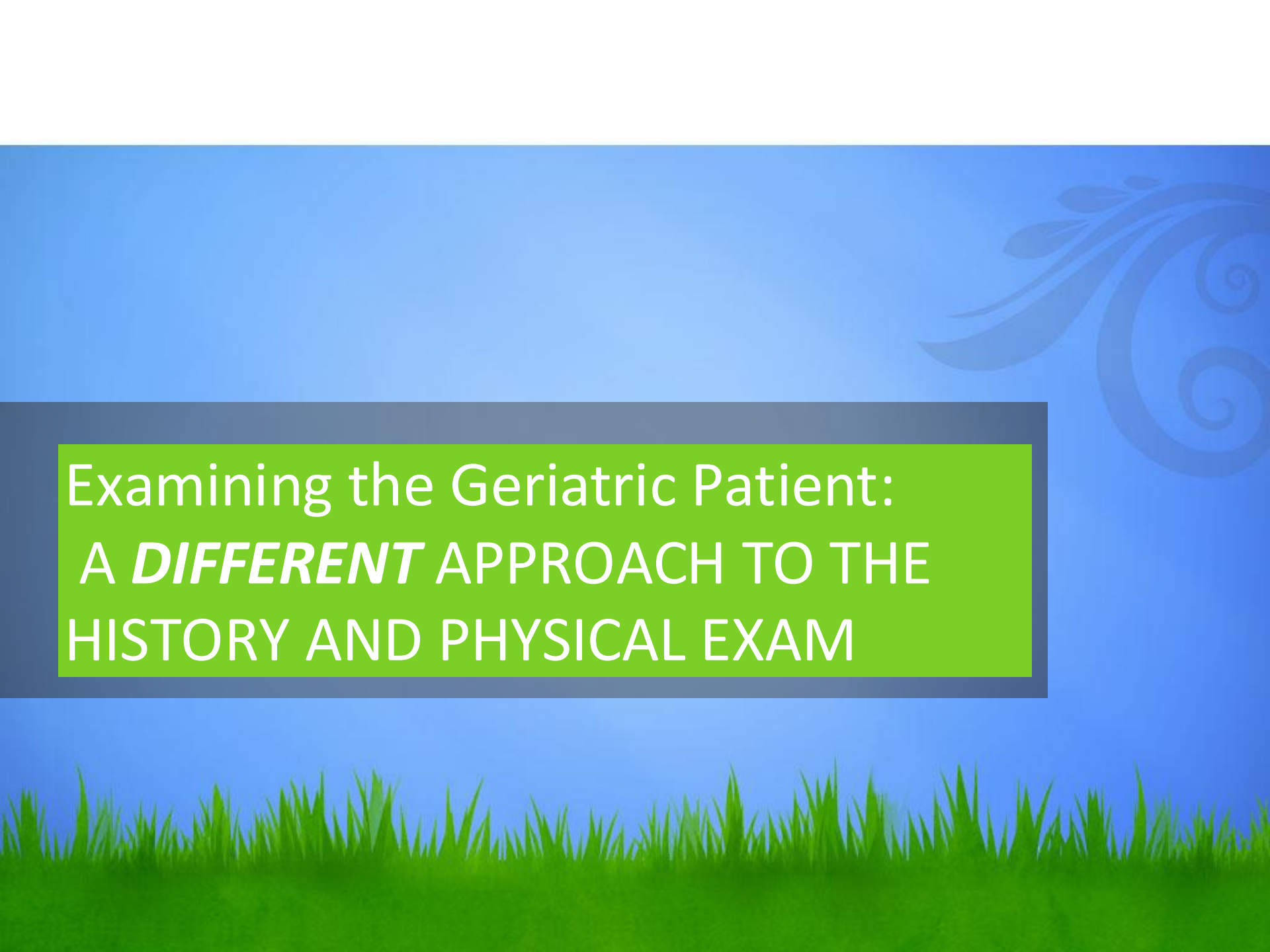
- Diagnostic testing as needed, but *consider*
- Patient's and family's wishes
- Cost
- Convenience/potential benefit or harm to patient
- Diagnostic/therapeutic yield
- Standard of care



CONSIDER INPUT FROM

- Caregivers
- Special Consultants
- Social workers
- Nurses, Home health
- Physical, Occupational, and Speech Therapists
- Nutrition consultations





Examining the Geriatric Patient: *A **DIFFERENT*** APPROACH TO THE HISTORY AND PHYSICAL EXAM

Some Differences in the History

1. Deficits Effect Communication

- Cognitive
- Affective
- Visual
- Auditory
- Mobility

2. Exam Visit Setting Matters More

- Quiet unhurried setting
- Family members
- Multiple visits

3. Chief Complaint vs. ROS

- HPI MAY BE vague and unclear
- ROS allows targeting geriatric syndromes

Some Differences in the Physical Examination

1. Physical deficits not reported may be revealed
2. Physical findings may be the result of a different disease
3. Physical findings may be attenuated by aging
4. Different high yield maneuvers
 - Postural blood pressure
 - Gait
 - Hearing
 - Visual acuity
 - Foot examination
 - Monitor weight



ORTHOSTATIC HYPOTENSION

- ORTHOSTATIC HYPOTENSION OCCURS IN 20% OF OLDER ADULTS AND UP TO 50% OF FRAIL NURSING HOME PATIENTS, ESPECIALLY WHEN THEY FIRST ARISE IN THE MORNING
- DEFINED AS A DROP IN SBP OF $\geq 20\text{mmHg}$ or DBP of $\geq 10\text{mmHg}$ WITHIN 3 MINUTES OF STANDING
- SYMPTOMS INCLUDE LIGHTHEADEDNESS, WEAKNESS, UNSTEADINESS, VISUAL BLURRING, AND IN 20-30% SYNCOPE.

PRESBYCUSIS age-related hearing loss

- The most common etiology of hearing loss in the elderly, appears to be most related to loss of inner-ear sensory structures.
- Prevalence as high as 80% in those over 80 yrs
- Results in significant impairment in quality of life, potentially leads to depression and social isolation, and may be further exacerbated by feelings of frustration that hearing loss can engender in caregivers.



PRESBYCUSIS

- Frequently misdiagnosed as cognitive dysfunction because the pt. may experience difficulties in understanding and communication.
- Subtle onset; begins as high-frequency hearing loss but progresses over time to encroach on frequencies in hearing people speak.



PRESBYCUSIS

- No one screening test has been shown to be superior to another.
- Hearing aids may improve communication abilities in older adults
- However, they require considerable patient motivation to be used effectively, and cost may be an issue.



- Visual impairment, often resulting from cataracts, macular degeneration, presbyopia, glaucoma, or diabetic retinopathy, is common in the elderly, but frequently underrecognized by patient and physician alike.
- Although visual acuity screening is inexpensive and easy to perform, it may be insufficient to adequately assess elderly patients.

Vision

- Although the USPSTF has found inconclusive evidence to justify routine screening in older adults, the American Academy of Ophthalmology recommends a comprehensive medical eye examination by an eye professional in asymptomatic adults age 65 years and older every 1-2 years; more frequent screening is recommended for patients with diabetes or glaucoma.

DEPRESSION IN THE ELDERLY

- Common in older adults, although under-recognized and undertreated.
- Older adults have higher rates of completed suicide than younger, although they attempt suicide less often.
- Depression should be suspected in the older adult if there are mood sx more intense than expected, poor response to medical tx, low motivation, and lack of engagement with providers



Depression in the elderly

- Screening tools include the Geriatric Depression Scale (GDS) or PHQ-9, both validated and have similar positive predictive values for the dx of depression.
- SSRI's are considered first-line pharmacotherapy for the elderly. Initiate therapy with a lower dose and increase as necessary.



WEIGHT LOSS

- IMPORTANT TO ASSESS NUTRITIONAL STATUS
- WHEN PATIENT'S PRESENT WITH $> 5\%$ WEIGHT LOSS IN 3 MONTHS OR $> 10\%$ DECREASE IN 6 MONTHS

Mind-Body-Spirit Connection

- William Sutherland observed:
“(sacral) sags and fascial drags lead to chronic rags”
– Contributions of Thought
- ASSSG (A Still Sutherland Study Group)
periodically dedicates a course to this topic
- American Academy of Osteopathy 2016
Convocation “Somatic Dysfunction and
Emotional Well-being: An Osteopathic
Approach to Mental Health

Reciprocal Tension Membrane – Dural Strain

- Sacral base anterior (postural) can drag the Primary Respiratory Mechanism into cranial extension

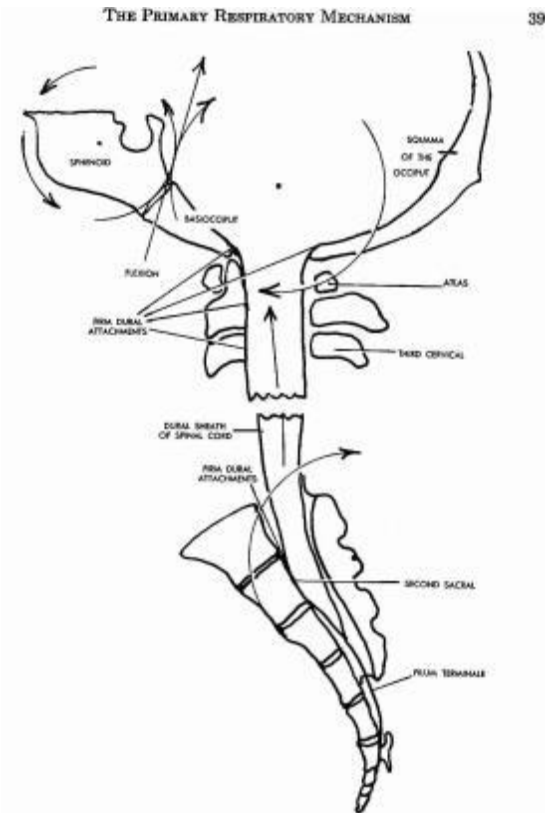


FIG. 20. THE CRANIOSACRAL MECHANISM IN FLEXION.

Many OMT Options

- Screen for somatic dysfunction
 - Decrease burden, improve homeostasis
- Treat to improve respiration, circulation
- Balance autonomic nervous system
 - Rib raising
 - Suboccipital release, sacral BLT
- Cranial
 - CV4
 - Venous sinus
 - Brain parenchyma



COGNITIVE FUNCTION

- Dementia affects more than 35% of persons age 90 years and older.
- Characterized by progressive decline in at least 2 cognitive domains (memory, attention, language, visuo-spatial, executive) that is severe enough to impair functioning.
- Mild cognitive impairment (MCI) represents cognitive deficits that do not affect daily functioning.
- MMSE has the strongest evidence base as a screening tool

ASSESSMENT OF THE OLDER DRIVER

- Involved in more fatal car crashes per mile driven than any other age group excepting those under 25yrs.
- A qualitative assessment that relies heavily on physician clinical judgment
- Family assessment of driving ability is more reliable than patient self assessment

Risk Factors for Impaired Driving

- Cognitive dysfunction
- Caregiver report of marginal or unsafe skills
- History of citations or accidents
- Driving < 60 miles/week
- Emotional aggression or impulsivity
- Use of ETOH and/or meds that affect the CNS
- Impaired mobility or coordination of neck or extremities
- Visual impairment
- Medical disorders that predispose to LOC

Impaired Drivers

- Drivers at highest risk should be counseled to “retire from driving”.
- A formal assessment by a driver rehab specialist (associated with hospital OT depts.) can be considered if the situation is unclear.
- Physician reporting responsibilities vary from state to state



Levels of Care

- Hospital setting for acutely ill elderly patients
- LTACH-Long-term acute care hospital for patients with on-going high intensity care needs who require regular physician input such as chronic mechanical ventilation, complex wound care needs or multiple intravenous therapies.
- Once the patient no longer requires high-acuity care, there are a multitude of care options available

- 
- Free-standing Rehab Hospitals provide intensive rehab services, defined as at least 3 hours per day, 5 days a week, from multiple therapy disciplines. Covered by medicare.
 - If patient is unable to tolerate that level of intensity, Skilled Nursing Facilities (SNF) provide rehab services. Medicare covers all or part of the cost of rehab in a SNF for 100 days, provided the patient has had a 3 day inpatient hospital stay prior to the SNF and makes continued progress during their stay.
- 

Levels of Care---Post-Hospital Rehabilitation

- Outpatient rehab is appropriate for patient's who are more functional and in some cases may be done at home.
- If the patient needs rehab services for a specific disease, such as cardiac rehab after an acute MI, or pulmonary rehab for chronic lung disease, this type of rehab is generally done in an outpatient setting.



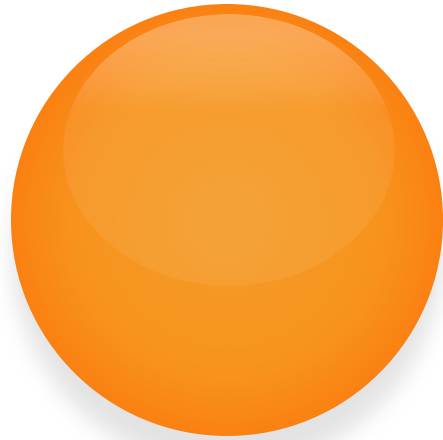
Other levels of care

- Nursing homes for patients who need substantial assistance with several ADL issues.
- Assisted living facilities
- In-home care
- Adult community day care facilities are helpful for patient's whose primary caregiver is at work or to provide caregiver respite.



CARE TRANSITIONS

- Complex patient's with many comorbid conditions often require care in more than one setting or may move between different settings, and **careful attention to these care transitions is essential.**
- Information that should be conveyed during transitions includes a summary of the hospital course, list of problems and diagnoses, baseline and discharge functional and cognitive status, reconciled medication list, outstanding test results, follow-up appts., information about goals of care and advanced directives. Checklists are helpful to assure completion.



FUNCTIONAL ASSESSMENT

COMPONENTS OF FUNCTIONAL STATUS



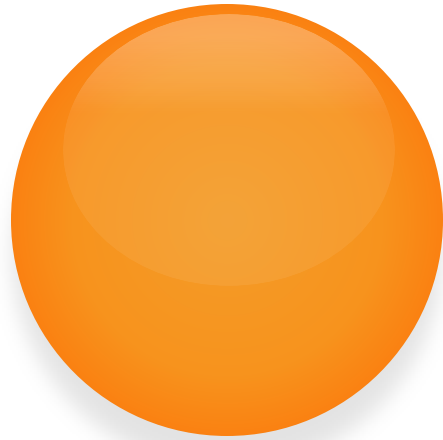
Functional Decline is multifactorial

- Factors that contribute to impaired functional status
 - Medical (disease)
 - Psychological
 - Social
 - Environmental
 - Aging (not disease)



SPECIAL SKILLS FOR EVALUATING FUNCTIONAL STATUS

- Mini Mental status exam (MMSE)
- Depression screen---PHQ-9 Questionnaire
- Activities of Daily Living (ADL) and Instrumental ADL (IADL) assessment
- Mobility/Fall assessment
- Nutrition screening



MOBILITY ASSESSMENT

GAIT , BALANCE , AND MUSCLE STRENGTH

Consequences of Falls

- Physical – decreased function or immobility
- Social – impacts quality of life
- Psychological – Fall-related fear & loss of self-efficacy
- Mortality



Fall Risk Factors

- Impairment in
 - Cognition
 - Vision
 - Balance and gait
 - Muscle strength
- Diseases, Symptoms
 - Arthritis
 - Parkinson's Disease
 - Depressive symptoms
 - Orthostatic Hypotension



Fall Risk Factors - Medication

- **Use of four or more medications is a risk factor**
- Medication Classes shown to have strongest link to an increased risk of falling
 - Benzodiazepines
 - Tricyclic antidepressants
 - Neuroleptic agents
 - SSRI's
 - Anticonvulsants
 - Class IA anti-arrhythmics (quinidine, procainamide and disopyramide)



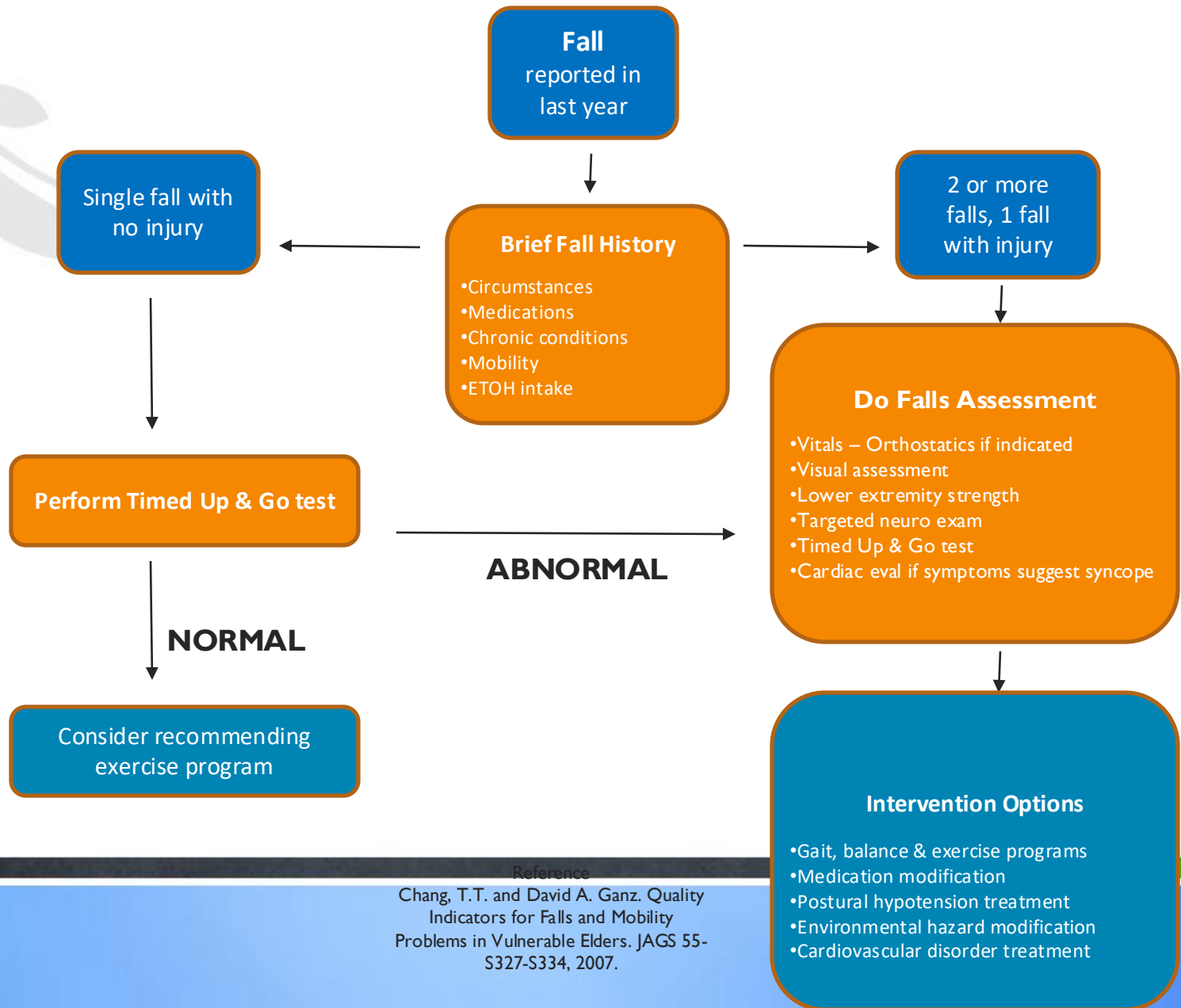
Physical Exam

- Check vitals – especially orthostatics
- Visual assessment
- Test for lower extremity strength
- Perform targeted neuro exam
 - proprioception, sensation
- Timed Get Up & Go Test
 - gait and balance abnormalities
 - normal <10 seconds
- Do cardiovascular work-up if syncopal episode



CASE

- 79 Y/O FEMALE PRESENTS FOR F/U
- HF, ARTHRITIS, DEPRESSION, DIFFICULTY SLEEPING
- MEDS: ANTIDEPRESSANT, DIURETIC, ACE-I, BETA-BLOCKER. ALSO TAKES OTC SLEEP AND ALLERGY MEDS
- CHRONIC CONDITIONS APPEAR STABLE
- DAUGHTER REPORTS 2 FALLS IN PAST 6 MO.



Reference
Chang, T.T. and David A. Ganz. Quality
Indicators for Falls and Mobility
Problems in Vulnerable Elders. JAGS 55-
S327-S334, 2007.

Intervention

- May require more than one intervention
- Gait, balance and exercise programs (OMM, PT, Tai Chi)
- Medication modification
- Postural hypotension treatment
- Environmental hazard modification
- Cardiovascular disorder treatment



ACTIVITIES OF DAILY LIVING (ADL'S) ASSESSMENT

The term ADL was first introduced by Sidney Katz in 1950. It's a collective term for all the basic skills you need in regular daily life. These include:

1. Ambulating. This includes the ability to walk, sit, stand, lie down and get up, and climb up and down stairs, both inside and outside your home.
2. Grooming. This includes all activities necessary to maintain [personal hygiene](#), like brushing your teeth, bathing, shaving, and hair and nail care.
3. Toileting. This includes the ability to control your bladder and bowels ([continence](#)), use the toilet safely, and clean yourself after use.
4. Dressing. This involves the ability to dress yourself properly, including using buttons and zippers.
5. Eating. This includes the ability to use cutlery and feed yourself.

ADLs essentially indicate your ability to care for yourself without assistance. According to a 2011 U.S. National Health Interview Survey, more than 20% of adults older than 85 years require assistance with ADL

Lawton Instrumental Activities of Daily Living Scale (IADL's)

IADLs include more complex activities that are essential to live independently. These are important activities that you don't need to do every day, unlike ADLs. They include:

Managing finances, such as paying bills and managing financial assets.

Managing transportation, either via driving or by organizing other means of transport.

Shopping and meal preparation. This covers everything required to get a meal on the table. It also covers shopping for clothing and other items required for daily life.

Housecleaning and home maintenance. This means cleaning kitchens after eating, keeping one's living space reasonably clean and tidy, and keeping up with home maintenance.

Managing communication, such as the telephone and mail.

Managing medications, which covers obtaining medications and taking them as directed.

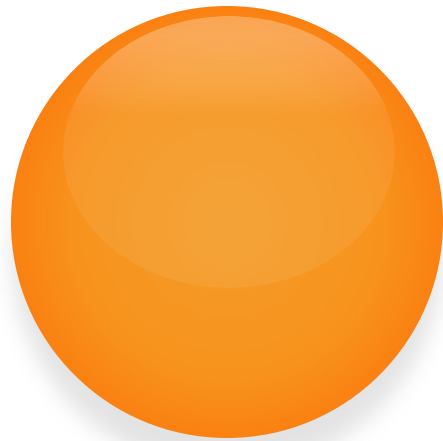
Because managing IADLs requires a fair amount of cognitive skill, it's common for IADLs to be affected when an older person is having difficulty with memory or thinking.

For those older adults who develop [Alzheimer's disease or a related dementia](#), IADLs will usually be affected before ADLs are.

The goal of the osteopathic physician is to find health

The ultimate goal of any health promotion intervention, is restoration and maintenance of the individual's FUNCTION, and not simply the treatment of disease.





MEDICATIONS

PHARMACOKINETIC and PHARMACODYNAMIC CHANGES IN THE GERIATRIC POPULATION



PHARMACOKINETIC CHANGES MOST AFFECTED BY THE AGING PROCESS

- **Distribution:** ↓ body water, ↓ lean mass
 - Reduction in loading dose for water-soluble substances (alcohol, aminoglycosides, digoxin)
- **Distribution:** ↑ body fat
 - Increased duration of effect of fat-soluble drugs (diazepam, amiodarone, haloperidol)
- **Metabolism: Reduced hepatic mass and hepatic blood flow**
- **EXCRETION: DECLINE IN RENAL FUNCTION (GFR,Tubular)**
 - Progressive decrease in renal excretion (e.g atenolol, digoxin)



DRUG ABSORPTION IS NOT QUANTITATIVELY AFFECTED BY AGING ITSELF WHY?

- ELDERLY PERSONS MAY HAVE DECREASED GI BLOOD FLOW AND INCREASED GASTRIC pH
- WHICH WOULD BE EXPECTED TO DECREASE DRUG ABSORPTION
- HOWEVER, DECREASED GI MOTILITY RESULTS IN LONGER TIME FOR DRUG ABSORPTION
- HENCE, **THE NET EFFECT ON ABSORPTION IS NEGLIGIBLE**



PHARMACODYNAMIC CHANGES WITH AGING

- RELATES TO THE PHYSIOLOGIC EFFECTS OF THE DRUG
- INCREASING AGE MAY RESULT IN AN INCREASED SENSITIVITY TO THE EFFECTS OF CERTAIN DRUGS
- EXAMPLES INCLUDE BENZODIAZEPINES AND OPIOIDS

Drug Interactions

- Drug-Drug
- Drug-Food
- Drug-Disease
- Drug-Herbal



Drug-Drug interactions: Absorption

- Cations may chelate antibiotics
 - quinolones, tetracyclines

Clinical Scenario

74 yo woman with osteoporosis and new onset dysuria. She takes alendronate weekly and daily calcium and vit D supplements. She begins ciprofloxacin for UTI, but calls you 3 days later when her symptoms have not improved.

Drug-Food Interactions: Metabolism

- CYP3A enzymes mediate the metabolism of Felodipine in the small intestines
- 250 ml of grapefruit juice may inhibit these enzymes for 24-38 hours resulting in increased bioavailability

Clinical Scenario

75 yo man whose HTN is well-controlled with felodipine. While attending a citrus convention in Orlando, he begins to feel lightheaded and fatigued. His blood pressure is noted to be 80/40.

Clinical Scenario

84 yo man with stable nocturia (2-3 x nightly) who develops a URI. He begins to take an over the counter “cold medicine”. 36 hours later, he goes to the emergency room for abdominal pain and inability to urinate.

Why? **Urinary retention**: anticholinergics (OTC cold medicine)

Drug-Disease Interactions

- **Urinary retention:** anticholinergics, tricyclics, alpha-agonists
- **Constipation:** anticholinergics, narcotics, calcium-channel blockers, tricyclics
- **Postural hypotension:** tricyclics, diuretics, antihypertensives
- **Falls:** benzodiazepines
- **Delirium:** anticholinergics, narcotics, and benzos
- **Exacerbation of Heart Failure:** NSAIDs
- **Urinary incontinence:** diuretics, alpha-blockers



PRESCRIBING CASCADES

- Occur when a **new** drug is prescribed to treat symptoms arising from an **unrecognized adverse drug event** related to an existing therapy.
- Patient is then at risk for developing additional adverse drug events
- Drug-induced symptoms in an older person can be easily misinterpreted as indicating a new disease or can be attributed to the aging process itself rather than the drug therapy

EXAMPLES OF PRESCRIBING CASCADES

- THIAZIDE DIURETIC RX > **ADVERSE DRUG EVENT** (ADE)=HYPERURICEMIA>ALLOPURINOL
- NSAID RX > **ADE**=B/P^ >ANTIHYPERTENSIVE RX
- ANTIPSYCHOTIC RX > **ADE**=EXTRAPYRAMIDAL SIGNS AND SYMPTOMS>ANTI-PARKINSONIAN THERAPY
- CHOLINESTERASE INHIBITOR RX FOR ALZHEIMER'S> **ADE**=URINARY INCONTINENCE> INCONTINENCE TREATMENT

PEARLS FOR PRESCRIBING AND MANAGING MEDICATIONS IN OLDER PERSONS

- Regular review of patient meds including non-prescription drugs and herbal supplements
- **Start low and go slow**
- Provide clear dosing instructions
- **REMEMBER TO RENAL DOSE** (May need a Lower dose or Longer intervals between doses)
- Longer periods between changes in doses
- Encourage the use of pillboxes



WHEN REVIEWING MEDICATION LISTS KEEP IN MIND THE FOLLOWING QUESTIONS

- Is each medication still indicated?
- Could any of the patient's complaints be from the medication?
- Are there any drug-drug or drug-disease interactions present?
- Is the benefit-to-risk ratio still the same as when the medication was started?
- Is non-drug therapy an option?
- Does the patient need additional education on the medication?



DIVERSE CHALLENGES

Geriatrics can be full of difficult but satisfying interactions—

From meeting people advanced in years but dynamic and engaged; to providing support with the transition at the very end of one's life.



TO CONCLUDE

- Aging is diverse, but thematic
- Develop and Rely on your fundamentals for evaluation and assessment
- Get familiar with geriatric related assessment tools
- Aim for a team approach as much as possible
- Strive for focused, stepwise interventions-
- Avoid complacency and delayed diagnoses
- Avoid overtreatment and polypharmacy



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